

Course Title - AI Literacy, Efficiency, and Ethics

Course Code: **IE423**

Instructors: **V Sunitha** and **Sanjay Srivastava**

Frequency: 2 lectures/Week (90 Minutes each). Lab Assignments to be done in the take-home mode

1. Course Overview

This course bridges the gap between technical capability and academic utility. The goal is *Metacognition*—using AI to "Learn how to Learn," conduct rigorous research, and prepare for an AI-driven job market.

2. Eligibility and Target Audience

This course is nominally offered for **credit** as an ICT elective for BTech 6th semester students. However, this course is also open to BTech 6th/8th semester, MTech, and PhD student for **audit**.

3. Evaluation Components

(a) In-class assignments / quizzes -	30%
(b) Take-home assignments –	30%
(c) Term Project and viva –	40%

Audit students are expected to perform at the minimum level of a CD grade in (a) and (b) to receive a PASS grade.

4. Required Resources

Software Stack (Institute Provided)

- **Text/Reasoning:** ChatGPT Team or Claude (Enterprise privacy).
- **Research:** Elicit or Consensus (Academic search).
- **Data:** Julius AI or Tableau (Visualization).
- **Accessibility:** Otter.ai (Transcription).

Reading List (Selected)

- *"The jagged frontier of AI"* (Harvard Business School Working Paper).
- *"Stochastic Parrots"* (Bender et al.).
- *"Bloom's Taxonomy Revisited"* (Oregon State University AI Guide).

5. Weekly Detailed Schedule

MODULE I: LITERACY & INCLUSION (The Foundation)

Goal: Moving from "Prompting" to "Algorithmic Thinking."

Week 1: The Paradigm Shift

- **Tue (Joint):** Predictive vs. Generative AI. The history from LSTMs to Transformers. Why "Chatting" is the wrong mental model.
- **Thu: The Ethics of Assistance.** Defining the "Gray Line" between Co-piloting (allowed) and Plagiarism (banned).

Week 2: Algorithmic Design Thinking

- **Tue (Joint):** It's not just about prompts; it's about **Decomposition**. Breaking complex problems into logic chains.
- **Thu: "Logic Mapping."** Mapping out a thesis argument's logic tree *before* asking AI to draft text.

Week 3: Inclusive Pedagogy (UDL)

- **Tue (Joint): AI as the Great Equalizer.** How AI supports neurodiversity (Dyslexia, ADHD, Language barriers).
- **Thu: "The Reader Bot."** Building a custom "Reader Assistant" that simplifies dense academic papers into bullet points/analogies.

Week 4: The Hallucination Problem

- **Tue (Joint): Probabilistic Reasoning.** Why AI lies? The concept of "Temperature" in models.
- **Thu: "Spot the Bot."** A gamified session where Faculty and Students compete to identify AI-generated text vs. human text, analyzing *tone* and *variance*.

MODULE II: RESEARCH & DATA FLUENCY (The Workflow)

Goal: Accelerating discovery without compromising rigor.

Week 5: The Literature Review

- **Tue (Joint): Synthesis vs. Search.** Using Semantic Scholar and Consensus.
- **Thu: "The Research Matrix."** Generating a comparative analysis table of 10 papers in 15 minutes to find research gaps.

Week 6: Data Storytelling

- **Tue (Joint): Visualizing Complexity.** Moving beyond static Excel charts to interactive dashboards.
- **Thu: "Thesis Viz."** Using AI to suggest 3 alternative ways (Heatmap, Scatter, Sankey) to visualize their project data.

Week 7: Code & Simulation (Guest: CS Dept)

- **Tue (Joint): AI in the Loop.** Understanding code generation even if you aren't a coder.
- **Thu: "Chat with Data."** Using Python libraries (via AI) to clean messy CSV datasets for their thesis.

Week 8: Mid-Semester Review

- **Tue (Joint): "Time-Saved" Showcase.** Lightning talks sharing the most efficient workflows discovered so far.
- **Thu:** Open Q&A and "Prompt Clinic" (Troubleshooting specific issues).

MODULE III: PEDAGOGY & CAREER (The Divergence)

Goal: Preparing for the future of work and assessment.

Week 9: Assessment Redesign

- **Tue (Joint): Bloom's Taxonomy 2.0.** Moving from "Remembering" to "Evaluating."
- **Thu: "Exam Prep."** Using AI to generate infinite variations of practice problems for upcoming finals.

Week 10: Personalized Learning

- **Tue (Joint): The Age of the Personal Tutor.** Adaptive learning systems.
- **Thu: "The Super-Tutor."** Training an AI to teach a complex concept (e.g., Quantum Mechanics) via Socratic questioning.

Week 11: The Future of Work

- **Tue (Joint): Human-AI Collaboration Models.** The "Centaur" vs. the "Cyborg" approach to work.
- **Thu: "Resume Optimization."** Beating the Applicant Tracking System (ATS) by optimizing keywords and formatting.

Week 12: Discipline Deep Dives

- **Tue (Joint): The Two Cultures.** Bridging STEM and Humanities in AI.
- **Thu: "The Shadow Portfolio."** Documenting their new AI skills (Prompting, Data Viz) to add to their LinkedIn/CV.

MODULE IV: ETHICS & POLICY (The Governance)

Goal: Establishing rules and understanding impact.

Week 13: Global Impact

- **Tue (Joint): The Hidden Cost.** Environmental impact (Water/Energy consumption of training) and Data Labor ethics.
- **Thu: "Debate Club."** AI & Intellectual Property: Who owns the output?

Week 14: Capstone Showcase

- **Tue (Joint):** Final Presentations.
- **Thu (Joint):** Final Presentations.

6. Certification Requirements

(AI Literacy Certificate)

To earn the completion certificate, students must submit:

1. **The Research Matrix:** An AI-generated literature review table.
2. **The Career Packet:** An AI-optimized Resume & Cover Letter.
3. **The Study Log:** Evidence of using AI for deep Socratic learning.